Statistically Gated AI Trading Infrastructure for Retail Derivatives

A Design, Statistical Judgement Layer and Pre-Registered Evaluation Protocol for a Solo Futures Trader on Zerodha Kite Connect

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Abstract. Regulatory studies reported by SEBI indicate that roughly nine in ten individual F&O traders lose money, so a retail trading system must be judged first by whether it can prove the absence of an edge cheaply and enforce discipline reliably. This paper presents the design of a six-component personal algorithmic trading system for Indian index and stock futures, built around one rule: artificial intelligence may make the system more cautious, never more aggressive. Our central contribution is a *statistical judgement layer*, a deterministic, lookahead-free module that sits between a rule-based setup and the risk gate and returns one of three verdicts (TAKE, DOWNSIZE, SKIP) with a size multiplier bounded above by one. The layer combines historical return distributions, volatility regimes, Wilson confidence intervals on win rate, bootstrap confidence intervals on after-cost expectancy, cross-position correlation, and drawdown-envelope monitoring. We specify every statistic, the verdict function, the walk-forward validation procedure, the phase gates that must pass before capital is risked, and a pre-registered protocol for comparing gated against ungated trading. **No live or backtested performance results are reported here;** this is a design and methodology paper, and its claims are limited to the properties of the design.

Keywords: algorithmic trading, derivatives, futures, backtesting, walk-forward validation, bootstrap, Wilson interval, risk management, SEBI retail algo framework, human-in-the-loop AI.

1. Introduction

Retail participation in Indian equity derivatives has grown rapidly, and so has the evidence of poor outcomes. SEBI's own studies, as summarised in the source material for this work, found that roughly nine in ten individual F&O traders lose money. A beginner who wants to use automation and AI to improve those odds faces two distinct problems. The first is epistemic: whether any rule-based setup has a positive expectancy after realistic costs. The second is behavioural: whether the trader will follow the rules once a loss occurs. A system that addresses only the first is incomplete, and a system that lets a language model place orders addresses neither.

We therefore treat the build as a research instrument. Most of its value arrives *before* any live trade: the backtest is expected to reject most candidate ideas, and discovering that fact for the price of a few thousand rupees of tooling is far cheaper than discovering it with margin money. The system's stated purpose is to enforce discipline and to test, honestly, whether an edge exists. It cannot promise profit.

Contributions. This paper makes four contributions:

- an architecture in which a single risk gate is the only path to an order, and AI is confined to roles that can only reduce risk (Section 3);

- a formal specification of a statistical judgement layer, including inputs, six statistics, a monotone verdict function and an audit schema (Section 4);
- a validation methodology combining an exact cost model, walk-forward testing, an overfitting diagnostic and pass/fail phase gates (Section 5);
- a pre-registered evaluation protocol for measuring whether the layer helps, written before any results exist (Section 7).

2. Regulatory and Operating Context

Personal algorithmic trading through a broker API is permitted, but the framework for retail algorithmic trading changed on 1 April 2026. The constraints below shape the system design directly; all were taken from broker and third-party explanatory pages and should be re-verified against Zerodha's current support pages before any execution code is written, because the rules have been extended and changed several times.

- **Static IP.** Orders are accepted only from an IP address registered with the broker. One third-party guide states that Zerodha permits up to three; this requires confirmation.
- **Daily authentication.** Continuous refresh sessions are no longer available, so the user logs in manually each trading morning. The system is therefore semi-automatic by design.
- **Order-rate threshold.** Below 10 orders per second, retail API users need no strategy registration; a solo system operates far below this.
- **Order types.** Brokers may convert market orders to price-protected orders. The system uses limit orders throughout.
- **Out of scope.** Running the system for others, or selling signals, falls under a different regime (exchange empanelment; Research Analyst registration for black-box strategies). This paper concerns personal use only.

Data constraint. Kite's historical API provides only daily candles for expired futures contracts. Intraday backtests therefore require minute data to be recorded from the start, or sourced elsewhere. Section 4 treats daily and intraday statistics separately for this reason.

3. System Architecture

The system comprises six small parts: a data store; a backtester that reproduces Zerodha's exact charges; one or two rule-based setups; a risk gate that sizes and vetoes every order; an approval queue in which the human confirms each trade; and a journal. The statistical judgement layer introduced in this paper is inserted between the setups and the risk gate (Figure 1).

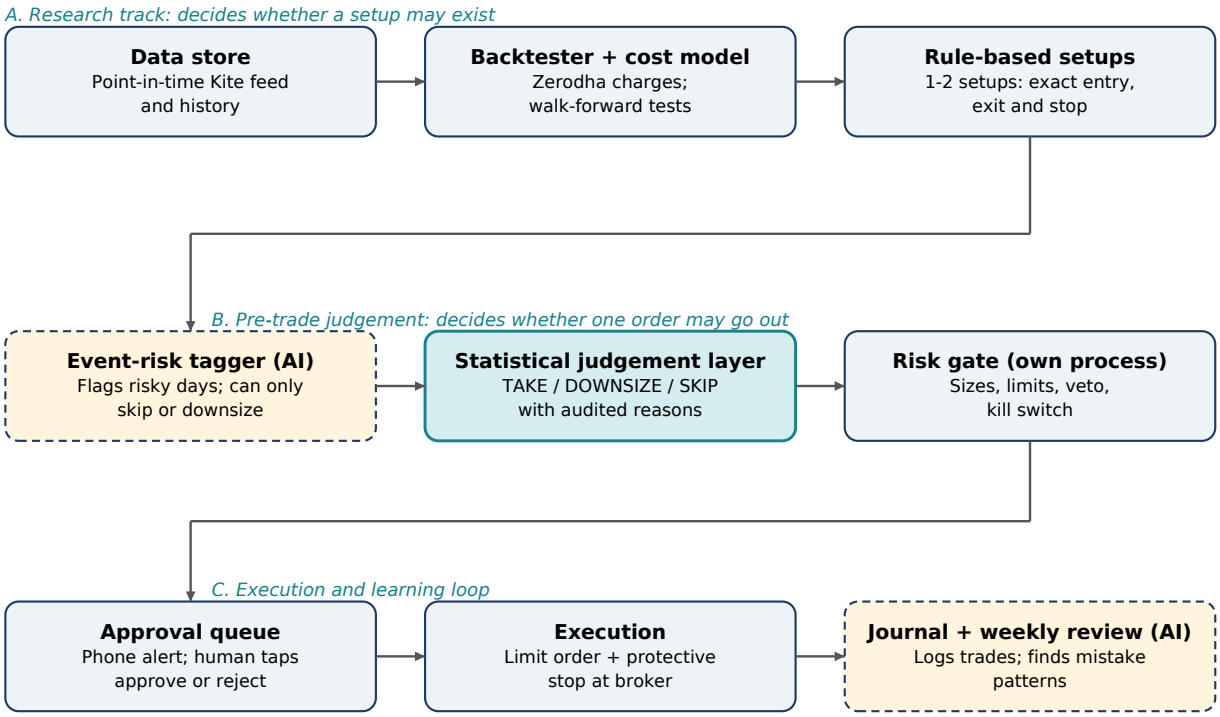


Figure 1. System architecture. Dashed amber boxes are the only places AI is used. The statistical judgement layer (teal) is deterministic code. Every order passes the risk gate, which runs as a separate process that signal code cannot override.

3.1 Design principles

- **One choke point.** Every order passes the risk gate. Position sizing and limits (daily loss, open positions, exposure, consecutive losses) live there and nowhere else.
- **Risk-reducing AI only.** AI may flag, downsize or veto a trade. It may never create or enlarge one.
- **Human approval.** The user may approve or reject a proposed trade but may not approve a trade the system did not propose.
- **Pause, do not tune.** If live results drift from the backtest, the setup is paused rather than adjusted. Parameters are never changed mid-week or after a losing streak.

3.2 Allocation of AI

Table 1 lists which jobs use AI and why. Language models are used where they are strong (reading text) and excluded wherever a decision must be reproducible and testable.

Job	AI?	Reason
Event-risk tagging (results, RBI, budget, global shocks)	Yes	Reading text suits language models. Output is a flag that can only skip or downsize.
Weekend journal review	Yes	Finds patterns in the trader's mistakes; the human approves any change, and changes re-enter the backtest.
Writing and testing code	Yes	A coding agent drafts modules; the human reviews and the tests decide.
Signal generation	No	Explicit statistical rules that can be backtested.
Statistical judgement	**No**	**Deterministic, tested, lookahead-free code; this paper's contribution.**
Position sizing and risk limits	No	One service decides every order.
Order placement	No	Nothing a model writes reaches the order path.
Parameter tuning	No	Walk-forward testing in code, not model opinion.

Table 1. Where AI belongs and where it stays out.

4. The Statistical Judgement Layer

The layer answers one question at the moment a setup fires: *given what was knowable at that instant, does the historical evidence justify this trade at full size, at reduced size, or not at all?* It is deterministic, contains no language-model calls, and can only return a verdict that reduces or preserves the size the setup would otherwise have.

4.1 Inputs and the point-in-time requirement

Let t denote the decision time and F_t the information available at t. The layer receives (i) point-in-time OHLCV history for the instrument (daily; minute data where recorded); (ii) the setup's historical trades from the backtester, each with entry, exit, stop and net profit after charges; (iii) the current market state; and (iv) the currently open positions. Every statistic S must satisfy

$$S(D) = S(D') \text{ for all datasets } D, D' \text{ that agree on } F_t, \tag{1}$$

that is, altering any row dated after t must leave the statistic unchanged. This property is enforced by an automated lookahead-bias test (Section 5.4). Missing, stale or incomplete data is never imputed: unknown means SKIP.

4.2 Historical distribution of outcomes

For each historical firing i of the setup, let r_i be the net return of the resulting trade after charges, and let $f_i(h)$ be the forward return over horizon h after the signal. The layer reports the mean, median, the 5th, 25th, 75th and 95th percentiles, skewness and excess kurtosis of $\{r_i\}$, and the percentile rank of today's conditioning variables (for example the current volatility level) within their historical distribution at firing. Heavy tails and negative skew matter because a high win rate can coexist with a negative expectancy when rare losses are large; the distribution is reported so that this is visible rather than assumed away.

4.3 Volatility regime

Realised volatility over a window of N daily log returns x_k is annualised as

$$\sigma_t = \sqrt{252} \cdot \sqrt{\left((1/(N-1)) \, \Sigma_{k=1}^{N} (x_k - \bar{x})^2 \right)}. \tag{2}$$

As a range-based alternative the layer also computes the Average True Range (Wilder, 1978) over 14 sessions. The percentile rank of the current volatility within a rolling window of W sessions is

$$\pi_t = (1/W) \cdot \#\{ s \in (t-W, t] : \sigma_s \le \sigma_t \}. \tag{3}$$

A regime label (low, normal, high, extreme) follows from π_t. Rather than hard-coding a regime switch, the layer converts π_t into a size multiplier through a linear ramp (Equation 4), so that behaviour changes smoothly and the thresholds can be audited.

$$\text{ramp}(x; a, b) = \min\{ 1, \max\{ 0, (b - x)/(b - a) \} \}. \tag{4}$$

Applied to volatility, $m_\sigma = \text{ramp}(\pi_t; \pi_{high}, \pi_{ext})$: the multiplier is 1 up to π_{high} and falls linearly to 0 at π_{ext}, so extreme volatility leads to SKIP.

4.4 Win rate with a confidence interval

A raw win rate from a modest number of trades is a noisy estimate. With w winning trades out of n and observed win rate $q = w/n$, the layer reports the Wilson (1927) score interval at confidence level $1 - \alpha$, where $z = z_{1-\alpha/2}$:

$$\left[q + z^2/2n \pm z\sqrt{\left(q(1-q)/n + z^2/4n^2 \right)} \right] / \left(1 + z^2/n \right). \tag{5}$$

The interval width, Δ_w, drives a multiplier $m_w = \mathrm{ramp}(\Delta_w; w_0, w_1)$: a wide interval means the win rate is poorly known and the trade is downsized. The Wilson interval is preferred to the normal approximation because it behaves well for small n and for proportions near 0 or 1.

4.5 Expected value after costs, with a bootstrap interval

Let g_i be the gross profit of trade i and c_i the total Zerodha charges (brokerage, exchange transaction charges, STT, stamp duty and GST) computed by the cost model that Section 5.1 requires to reproduce a real contract note to the rupee. The after-cost expectancy per trade and its decomposition are

$$E = (1/n)\,\Sigma_i\,(g_i - c_i) = q \cdot W - (1-q) \cdot L, \quad b = W/L, \tag{6}$$

where W and L are the mean net win and mean absolute net loss, and b is the payoff ratio. Because trade outcomes are skewed and few, the layer avoids parametric assumptions and builds a percentile bootstrap interval (Efron, 1979): resample the n net outcomes with replacement B times (default 10,000, fixed seed for reproducibility), compute E^*_j for each resample, and take the $\alpha/2$ and $1-\alpha/2$ empirical quantiles as $[E_{\mathrm{lo}}, E_{\mathrm{hi}}]$. The decision rule is deliberately conservative: it keys on the *lower* bound, not the point estimate. A setup whose $E_{\mathrm{lo}} \le 0$ has not demonstrated a positive after-cost edge and is not traded.

4.6 Correlation with open positions

Two positions that look diversified can behave as one. For the proposed instrument with return series x and each open position j with series y_j, the layer computes the rolling Pearson correlation over L sessions and takes the largest value among positions with the same directional exposure:

$$\rho_{\max} = \max_j \rho_L(x, y_j), \quad m_\rho = \mathrm{ramp}(\rho_{\max}; \rho_0, \rho_1). \tag{7}$$

Correlation is itself unstable in stressed markets, which is when it matters most; the layer therefore reports the window length with the value and treats a high correlation as a reason to reduce, never to increase, exposure.

4.7 Drawdown envelope

Let V_t be the cumulative net equity curve of the setup and $M_t = \max_{s \le t} V_s$ its running peak. The drawdown is $\mathrm{DD}_t = (M_t - V_t)/M_t$. Let $\mathrm{MaxDD}_{\mathrm{train}}$ be the maximum drawdown in the training window. The layer compares the current drawdown with that historical envelope:

$$m_d = \mathrm{ramp}(\,\mathrm{DD}_t \,/\, \mathrm{MaxDD}_{\mathrm{train}}\,;\, \kappa_0, \kappa_1\,). \tag{8}$$

A setup running well outside the drawdown it exhibited in testing is behaving differently from the process that was validated, which is the signature of decay or regime change. Consistent with Section 3.1, the response is to shrink and pause, not to re-tune.

4.8 The verdict function

The six statistics are combined by a rule that cannot increase size. Hard failures produce SKIP directly:

- fewer than $n_{\min}$ historical trades (default 200);
- $E_{\mathrm{lo}} \le 0$ after costs;
- data missing, stale or incomplete at decision time;
- out-of-sample degradation below threshold (Section 5.3).

Otherwise the size multiplier is the minimum of the individual multipliers, and the verdict follows from its value:

$$m = \min\{\ 1,\ m_{\sigma},\ m_{w},\ m_{\rho},\ m_{d}\ \}. \tag{9}$$

The verdict is SKIP if $m < m_{\min}$, TAKE if $m = 1$, and DOWNSIZE otherwise.

The final order size is m times the size the risk gate would permit, and the risk gate may reduce it further. Because $0 \le m \le 1$ by construction, the layer is *monotone safe*: no input, however favourable, can make a trade larger than it would have been without the layer. Using the minimum rather than the product prevents several mild concerns from compounding into an arbitrarily tiny position, while still letting any single severe concern bind. The floor $m_{\min}$ (default 0.25) avoids dust-sized orders whose charges would dominate their expected return.

4.9 Configuration, output and audit

No threshold is embedded in code. All parameters live in a versioned configuration file; Table 2 lists them with *illustrative* defaults. These defaults are starting points, not findings: they are to be fixed using training windows only and frozen before any out-of-sample or paper-trading evaluation.

Parameter	Symbol	Illustrative default
Minimum historical trades	$n_{\min}$	200
Confidence level / bootstrap resamples	$1 - \alpha$ / B	95% / 10,000 (fixed seed)
Volatility window; ramp thresholds (high, extreme)	W; π_{high}, π_{ext}	252 sessions; 0.80, 0.95
Wilson width ramp	w_0, w_1	0.10, 0.25
Correlation window; ramp thresholds	L; ρ_0, ρ_1	60 sessions; 0.50, 0.90
Drawdown-envelope ramp	κ_0, κ_1	0.8, 1.2
Out-of-sample degradation threshold	δ	0.5
Minimum size multiplier	$m_{\min}$	0.25

Table 2. Configurable parameters of the statistical judgement layer (illustrative defaults, to be set on training data only).

Each call returns a serialisable record containing the verdict, the multiplier, every statistic with its value, the list of reasons (each reason paired with the number that triggered it), the data-as-of timestamp and the configuration version. Every record is written to the journal, so that any past decision can be reconstructed and challenged. An illustrative record, in schematic form:

```
verdict: DOWNSIZE multiplier: 0.47 data_as_of: 2026-10-15T09:10+05:30 config: v1.3
reasons:
  volatility percentile 0.88 (ramp 0.80→0.95) → m_sigma = 0.47
  max correlation with open positions 0.71 → m_rho = 0.48
  E_lo > 0, n ≥ n_min, win-rate interval width acceptable → no hard failure
  m = min(1, 0.47, 1.0, 0.48, 1.0) = 0.47 → floor test passed
```

The record above is a format illustration with invented numbers, not an observation.

5. Validation Methodology

5.1 Cost model fidelity

Futures strategies have thin per-trade margins, so an inexact cost model can manufacture or erase an edge. The cost model is accepted only when it reproduces two to three months of real contract notes *to the rupee*. Slippage versus the backtest is recorded for every live trade thereafter.

5.2 Walk-forward evaluation

Data are divided into successive windows. For window k, all statistics and thresholds are computed on the training segment $[t_{k-1}, t_k)$ and applied, frozen, to the test segment $[t_k, t_{k+1})$. The windows then roll forward. The layer's verdicts on every test segment are generated exactly as they would have been in real time.

5.3 Overfitting diagnostic

Backtest overfitting is the most likely way for a retail system to fail (Bailey et al., 2017). The layer reports in-sample and out-of-sample expectancy side by side and computes the degradation ratio

$$D = E_{\mathrm{OOS}} / E_{\mathrm{IS}}, \quad \text{flag possible overfitting if } D < \delta. \tag{10}$$

A single parameter-light setup, a minimum sample of 200 trades, and a strict separation of training and testing data reduce, but do not remove, this risk. The ratio is a warning signal, not a proof, and a flagged setup is returned to research rather than retuned in place.

5.4 Automated tests

- **Lookahead test.** Alter rows after t; no statistic may change (Equation 1).
- **Known-answer tests.** A coin-flip setup with zero expectancy must produce SKIP; a synthetic setup with a strong, stable edge must produce TAKE.
- **Monotonicity test.** For randomly generated inputs, the multiplier never exceeds 1 and the output size never exceeds the risk gate's input size.
- **Missing-data test.** Injecting NaNs, gaps or stale timestamps must yield SKIP, never a guessed value.

5.5 Phase gates

Development proceeds in phases. Each ends at a pass/fail gate, and a failed gate sends the work back rather than forward (Table 3). The weeks shown are a part-time plan, not a promise.

Phase	Timing	Work	Gate to pass
0 Setup	Week 1	Accounts, static IP, API app	Authenticated API call succeeds from the whitelisted IP
1 Data and costs	Weeks 2–4	Data store, Zerodha charge model	Cost model reproduces a real contract note to the rupee
2 Backtest one setup	Weeks 4–8	Walk-forward and out-of-sample tests; statistical layer	Positive after costs over 200+ trades and holds out-of-sample
3 Risk gate and tools	Weeks 8–10	Gate, approval queue, journal	Gate blocks a bad order; kill switch fires in testing
4 Paper trading	Weeks 10–22	Live feed, no real orders	Paper results stay within tolerance of the backtest
5 Live, one lot	Week 22 on	Approval-gated, minimum size	100 live trades match the backtest before any size increase
6 Scale	After gate 5	Small steps only	n/a

Table 3. Build roadmap and pass/fail gates.

6. Feasibility and Cost

Running costs are estimated at roughly Rs 2,800–6,500 per month, dominated by the Kite Connect subscription, a fixed IP address and LLM usage; all other software is open source. The build is estimated at about 10 weeks part-time at 10–15 hours a week, followed by 8–12 weeks of paper trading. Several items are the author's estimates or come from third-party pages and have not been independently verified (Table 4).

Item	Rs / month	Basis
Kite Connect base subscription (historical data included)	2,000	Zerodha support page; the historical-data add-on fee was reported dropped from 8 Feb 2025 (Kite forum)
Static IP from an ISP	~200–600	Third-party guide; varies by ISP; unnecessary on a cloud server
Cloud server with fixed IP (Mumbai region)	~500–1,500	Author's estimate; check current prices
LLM API (event tagger + weekly review)	~500–3,000	Author's estimate; scales with events tagged
Python, Postgres/DuckDB, vectorbt, Telegram bot	0	Open source
Total	**~2,800–6,500**	Low: ISP static IP. High: cloud server and heavy LLM use

Table 4. Estimated monthly running costs.

Why fixed cost matters. Fixed tooling cost is a hurdle rate. At Rs 4,000 a month, an account of Rs 2,00,000 must earn 2% a month merely to cover tooling, before brokerage, exchange charges, STT and tax (illustrative figures). For small accounts this alone can erase a modest edge, and it is a further reason the layer requires $E_{\text{lo}} > 0$ *after* costs rather than on gross returns. Trading capital is separate from tooling cost, and per-lot margin should be checked on Zerodha's margin calculator first.

7. Pre-Registered Evaluation Protocol

To avoid fitting the evaluation to the results, the comparisons below are fixed before any data are examined. The unit of comparison is the same setup traded (a) *ungated*, with the risk gate only, and (b) *gated*, with the statistical judgement layer in front of the risk gate, on identical out-of-sample and paper-trading periods.

7.1 Hypotheses

- **H1 (risk).** Gating reduces maximum drawdown relative to the ungated variant over the same out-of-sample period.
- **H2 (cost of caution).** Gating does not reduce after-cost expectancy *per unit of risk taken* by more than a pre-set tolerance, so any gain in H1 is not bought purely by trading less.
- **H3 (stability).** The SKIP and DOWNSIZE rates in paper trading stay within tolerance of their rates in the walk-forward test; a large shift signals drift.

7.2 Metrics

After-cost expectancy per trade with its bootstrap interval; maximum drawdown and time under water; profit factor; trade count; SKIP, DOWNSIZE and TAKE frequencies; slippage versus backtest; and cost drag as the fraction of gross profit consumed by charges. All metrics are reported with intervals, never as bare point estimates.

7.3 Rules of interpretation

- A hypothesis is judged on out-of-sample or paper-trading data only. In-sample improvement is not evidence.
- Null or negative results are reported with the same prominence as positive ones. If no setup survives Phase 2, that is the primary finding, not a failure of the project.
- Thresholds in Table 2 are frozen before evaluation. Any later change restarts the evaluation clock.
- With the small samples typical of a single retail setup, the intervals will be wide; a result that depends on the point estimate alone is treated as inconclusive.

8. Risks and Limitations

The largest risks are, in order: an overfit backtest, leverage, and the trader overriding the system; technology failures come after these.

Risk	Mitigation in the design	Residual limitation
Overfitting	Few parameters; walk-forward and out-of-sample tests; degradation ratio δ; 200+ trade minimum	Multiple trials across ideas still inflate false discoveries; the ratio is a warning, not proof
Costs and slippage	Cost model matched to contract notes; slippage tracked live	Slippage in stressed markets may exceed anything seen in testing
Leverage	Fixed risk per trade; hard daily loss limit in the separate risk gate	Gap moves can exceed stop levels
Outages	Protective stop placed at the broker on entry; reconciliation every few minutes	Expired login or changed IP with an open position remains operationally dangerous
Rule changes	Re-check SEBI and Zerodha pages before going live	The regime has changed repeatedly and may change again
The trader	Approval queue; no unproposed trades; no mid-week parameter changes	Cannot be removed, only made harder; a beginner's strongest failure mode is overriding the gate after a loss
Non-stationarity	Drawdown-envelope monitor; pause rather than tune	Past distributions may not describe future markets, and every statistic here is backward-looking

Table 5. Principal risks and residual limitations.

Further limitations apply to this paper itself. First, it reports no empirical results; the layer's usefulness is a hypothesis to be tested under Section 7. Second, the six statistics are standard and individually well understood, but their combination by a minimum-of-ramps rule is a design choice, not a derived optimum. Third, rolling correlation and volatility percentiles are sensitive to window choice. Fourth, the regulatory and cost figures were read from search excerpts of third-party and broker pages on 13 October 2026 and were not independently verified. Finally, a system built for one person's personal trading cannot be generalised to a service for others without a different legal and compliance analysis.

9. Conclusion

We have described a retail derivatives trading system whose guiding principle is that AI may only make it more cautious, and we specified a statistical judgement layer that turns that principle into testable code. The layer evaluates six properties of a setup (return distribution, volatility regime, win-rate uncertainty, after-cost expectancy with a bootstrap interval, correlation with open positions, and drawdown behaviour) and returns TAKE, DOWNSIZE or SKIP with a multiplier that can never exceed one. Its design is backed by lookahead and known-answer tests, walk-forward validation, phase gates and an audit trail. The honest expected outcome of the first months is not profit but information: either a setup survives realistic costs and out-of-sample testing, or the system has saved its owner from trading a setup that would not have. Both outcomes are results. Future work is to implement the layer, execute Phases 0–4, and report the evaluation in Section 7 exactly as pre-registered.

Appendix A. Pseudocode for the verdict function

```
function judge(setup, t, config):
  D = point_in_time_data(t) # no rows after t
  if D is stale or incomplete: return SKIP(reason="data")
  trades = backtest_trades(setup, up_to=t)
  if len(trades) < n_min: return SKIP(reason="sample")
  E, E_lo, E_hi = expectancy_bootstrap(trades, B, alpha, seed)
  if E_lo <= 0: return SKIP(reason="no after-cost edge")
  if degradation(trades) < delta: return SKIP(reason="overfit warning")
  m_sigma = ramp(vol_percentile(D), pi_high, pi_ext)
  m_w     = ramp(wilson_width(trades), w0, w1)
  m_rho   = ramp(max_corr(D, open_positions), rho0, rho1)
  m_d     = ramp(drawdown_ratio(trades), kappa0, kappa1)
  m = min(1, m_sigma, m_w, m_rho, m_d)
  verdict = SKIP if m < m_min else (TAKE if m == 1 else DOWNSIZE)
  log_to_journal(verdict, m, all_statistics, reasons, t, config.version)
  return verdict, m
```

Appendix B. Pre-live checklist

Live trading begins only when every item holds:

- [ ] The cost model reproduces a real contract note to the rupee.
- [ ] After-cost positive expectancy over 200+ backtest trades, holding out-of-sample.
- [ ] The risk gate runs as a separate process and blocks a deliberately bad order in testing.
- [ ] The kill switch is tested for daily loss limit, consecutive losses and abnormal slippage.
- [ ] Paper trading on the live feed for 8–12 weeks, within tolerance of the backtest.
- [ ] The static IP is whitelisted and a full morning login has been rehearsed.
- [ ] The first live phase is one lot, approval-gated.

References

[1] Bailey, D. H., Borwein, J. M., López de Prado, M., and Zhu, Q. J. (2017). The probability of backtest overfitting. *Journal of Computational Finance*, 20(4).

[2] Efron, B. (1979). Bootstrap methods: another look at the jackknife. *The Annals of Statistics*, 7(1), 1–26.

[3] Wilder, J. W. (1978). *New Concepts in Technical Trading Systems*. Trend Research.

[4] Wilson, E. B. (1927). Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158), 209–212.

[5] SEBI studies on the profitability of individual F&O traders, as cited in the source build guide (13 October 2026). Primary documents were not re-checked for this draft.

[6] Zerodha support: Kite API charges.
support.zerodha.com/category/trading-and-markets/kite-api/articles/what-are-the-charges-for-kite-apis

[7] Kite forum: historical data now included. kite.trade/forum/discussion/comment/49421/

[8] Zerodha: explaining the latest SEBI algo trading regulations.
zerodha.com/z-connect/featured/explaining-the-latest-sebi-algo-trading-regulations

[9] Tradejini: what SEBI's new algo trading rules mean for you.
tradejini.com/blogs/what-sebis-new-algo-trading-rules-mean-for-you

[10] FYERS: notice on the new SEBI framework for retail algo trading from 1 April 2026.
fyers.in/notice-board/new-sebi-framework-for-retail-algo-trading-from-april-01-2026/

[11] DEV Community: SEBI 2026 static IP mandate, complete compliance guide for retail traders.
dev.to/shaktitiwari/sebi-2026-static-ip-mandate-complete-compliance-guide-for-retail-traders-25ba

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